

Example 1

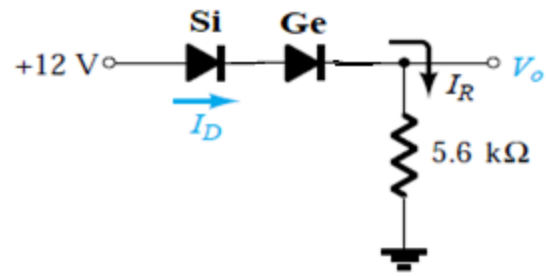
Determine the output voltage V_o and current I_D in the series circuit shown.

Solution

$$\bullet \quad 12 - 0.7 - 0.3 - V_o = 0 \quad \text{KVL}$$

$$V_o = 12 - 0.7 - 0.3 \Rightarrow V_o = 11 \text{ V}$$

$$\bullet \quad I_D = \frac{V_o}{R} = \frac{11}{5.6 \text{ k}} = 1.96 \text{ mA}$$

**Example 2**

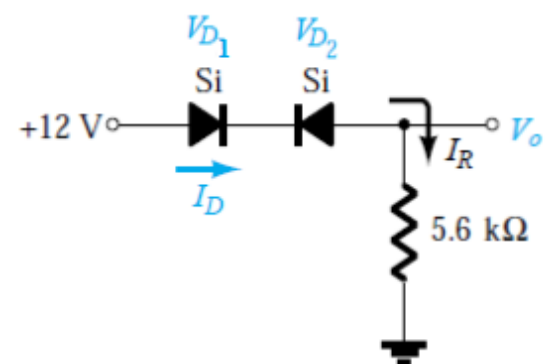
Determine the output voltage V_o , I_D , V_{D1} and V_{D2} in the series circuit shown.

$$\triangleright D_1 \text{ is on but } D_2 \text{ is off} \Rightarrow I_D = 0 \text{ and } V_o = 0$$

$$12 - V_{D1} - V_{D2} - V_o = 0 \quad \text{KVL}$$

$$V_{D2} = 12 - V_{D1} - V_o \Rightarrow V_{D2} = 12 - 0 - 0 = 11 \text{ V}$$

Note: For practical diode, when $I_D = 0 \Rightarrow V_D = 0$

**Example 3**

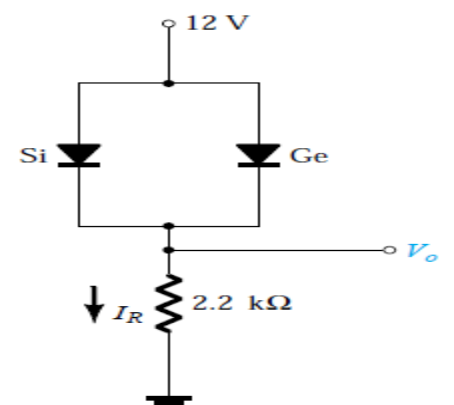
Determine the output voltage V_o and current I_R in the series circuit shown.

Solution

$$\bullet \quad 12 - 0.3 - V_o = 0 \quad \text{KVL}$$

$$V_o = 12 - 0.3 \Rightarrow V_o = 11.7 \text{ V}$$

$$\bullet \quad I_R = \frac{V_o}{R} = \frac{11.7}{2.2 \text{ k}} = 5.32 \text{ mA}$$

**Example 4**

Determine V_{o1} , V_{o2} , and I for the network shown.

Solution

$$V_{o1} = 0.7 \text{ V and } V_{o2} = 0.3 \text{ V}$$

I_1 through $R_1 = 1 \text{ k}\Omega$:

$$\bullet \quad 20 - I_1 R_1 - V_{o1} = 0 \quad \text{KVL}$$

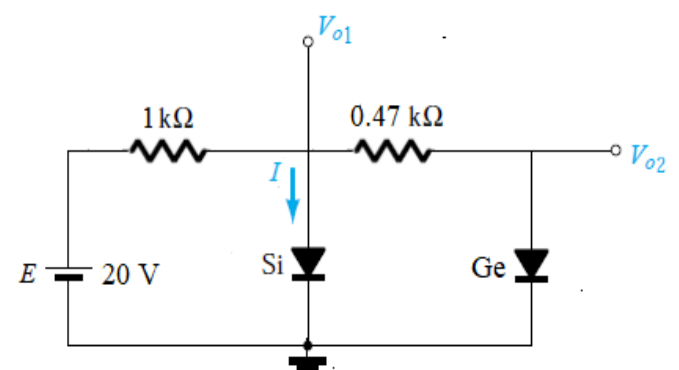
$$I_1 = \frac{20 - V_{o1}}{R_1} = \frac{20 - 0.7}{1 \text{ k}} = 19.3 \text{ mA}$$

I_2 through $R_2 = 0.47 \text{ k}\Omega$:

$$\bullet \quad V_{o1} - I_2 R_2 - V_{o2} = 0 \quad \text{KVL}$$

$$I_2 = \frac{V_{o1} - V_{o2}}{R_2} = \frac{0.7 - 0.3}{0.47 \text{ k}} = 0.85 \text{ mA}$$

$$\bullet \quad I = I_1 - I_2 = 19.3 - 0.85 = 18.45 \text{ mA}$$



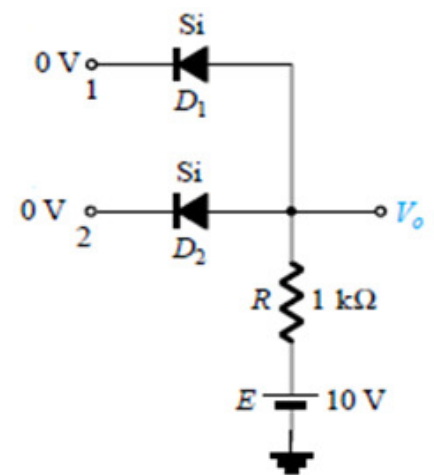
Example 5

Determine V_o for the network shown with 0 V on both inputs.

Solution

$$\bullet \quad 0 - (-0.7) - V_o = 0 \quad \text{KVL}$$

$$0 + 0.7 - V_o = 0 \Rightarrow V_o = 0.7 \text{ V}$$

**Example 6**

Determine I_R , V_1 , V_2 and V_o for the network shown

$$10 - I_R R_1 - V_D - I_R R_2 + 5 = 0 \quad \text{KVL}$$

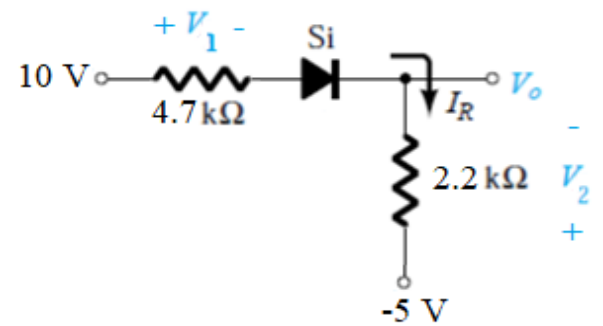
$$I_R = \frac{10 - V_D + 5}{R_1 + R_2} = \frac{10 - 0.7 + 5}{(4.7 + 2.2)k} = 2.072 \text{ mA}$$

- $V_1 = I_R R_1 = 2.072 \text{ m} \times 4.7 \text{ k} = 9.74 \text{ V}$
- $V_2 = -I_R R_2 = -2.072 \text{ m} \times 2.2 \text{ k} = -4.56 \text{ V}$

$$V_o + V_2 + 5 = 0 \Rightarrow V_o = -V_2 - 5 = -(-4.56) - 5 = -0.44 \text{ V}$$

OR

$$10 - V_1 - V_D - V_o = 0 \Rightarrow V_o = 10 - V_1 - V_D = 10 - 9.74 - 0.7 = -0.44 \text{ V}$$



Example 1

For the circuit shown below. Find

- i) the output voltage
- ii) the voltage drop across series resistance
- iii) the current through zener diode.

Solution

Check

$$I_L = \frac{V_i}{R_S + R_L} = \frac{120}{5 + 10} = 8 \text{ mA}$$

$$V_L = I_L \times R_L = 8 \text{ m} \times 10 \text{ k} = 80 \text{ V}$$

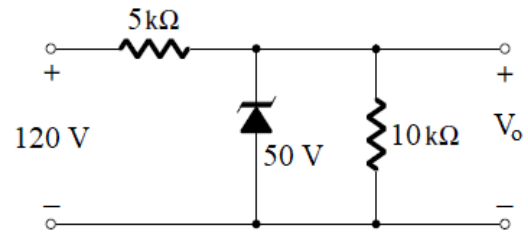
$$V_L > V_Z \Rightarrow \text{Zener is on } V_L = V_Z = 50 \text{ V}$$

$$\text{i) } V_o = 50 \text{ V}$$

$$\text{ii) } V_R = V_i - V_Z = 120 - 50 = 70 \text{ V}$$

$$\text{iii) } I_L = \frac{V_L}{R_L} = \frac{50}{10} = 5 \text{ mA} \quad \text{and} \quad I_R = \frac{V_R}{R} = \frac{70}{5} = 14 \text{ mA}$$

$$I_Z = I_R - I_L = 14 - 5 = 9 \text{ mA}$$



Example 2

The zener diode shown below has $V_Z = 18 \text{ V}$. The voltage across the load stays at 18 V as long as I_Z is maintained between 200 mA and 2 A. Find the value of series resistance R so that V_o remains 18 V while input voltage V_i is free to vary between 22 V to 28V.

Solution

$$I_L = \frac{V_L}{R_L} = \frac{18}{18} = 1 \text{ A}$$

$$I_R = I_{Z \max} + I_L = 2 + 1 = 3 \text{ A}$$

$$V_R = V_{i \max} - V_Z = 28 - 18 = 10 \text{ V}$$

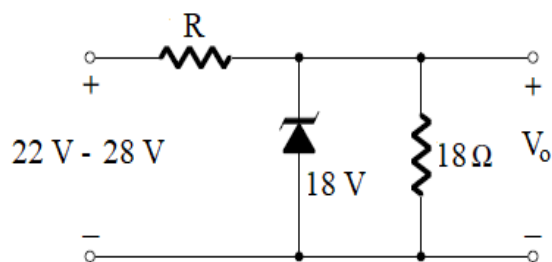
$$I_R = \frac{V_R}{R} \Rightarrow R = \frac{V_R}{I_R} = \frac{10}{3} = 3.33 \Omega$$

OR

$$I_R = I_{Z \min} + I_L = 0.2 + 1 = 1.2 \text{ A}$$

$$V_R = V_{i \min} - V_Z = 22 - 18 = 4 \text{ V}$$

$$I_R = \frac{V_R}{R} \Rightarrow R = \frac{V_R}{I_R} = \frac{4}{1.2} = 3.33 \Omega$$



Example 3

What value of series resistance is required when three 10 W, 10 V, 1000 mA zener diodes are connected in series to obtain a 30 V regulated output at no load from a 45 V d.c. power source ?

Solution

- $V_Z = 10 + 10 + 10 = 30 \text{ V}$
- $I_Z = 1000 \text{ mA}$, $V_o = 30 \text{ V}$, $V_i = 45 \text{ V}$

$$I_R = I_Z + I_L = 1000 + 0 = 1000 \text{ mA} = 1 \text{ A}$$

$$V_R = V_i - V_Z = 45 - 30 = 15 \text{ V}$$

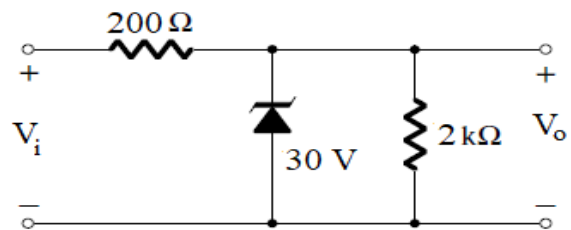
$$I_R = \frac{V_R}{R} \Rightarrow R = \frac{V_R}{I_R} = \frac{15}{1} = 15 \Omega$$

Example 4

Over what range of input voltage will the zener circuit shown below maintain 30 V across 2000 Ω load, assuming that series resistance $R = 200 \Omega$ and zener current rating is 25 mA ?

Solution

- $R = 200 \Omega = 0.2 \text{ k}\Omega$
- $I_L = \frac{V_L}{R_L} = \frac{30}{2} = 15 \text{ mA}$



For minimum input voltage $\Rightarrow I_{Z \text{ min}} = 0$:

$$I_R = I_{Z \text{ min}} + I_L = 0 + 15 = 15 \text{ mA}$$

$$V_R = I_R \times R = 15 \times 0.2 = 3 \text{ V}$$

$$V_{i \text{ min}} = V_R + V_Z = 3 + 30 = 33 \text{ V}$$

For maximum input voltage $\Rightarrow I_{Z \text{ max}} = 25 \text{ mA}$:

$$I_R = I_{Z \text{ max}} + I_L = 25 + 15 = 40 \text{ mA}$$

$$V_R = I_R \times R = 40 \times 0.2 = 8 \text{ V}$$

$$V_{i \text{ min}} = V_R + V_Z = 8 + 30 = 38 \text{ V}$$

Voltage input range = from 33 V to 38 V

Example 5

Design a network to maintain load voltage at 12 V for a load variation current from 0 to 200 mA when the input voltage is 16 V.

Solution

- $V_i = 16 \text{ V}, V_L = V_Z = 12 \text{ V}$

For maximum load current $\Rightarrow I_{Z \min} = 0$ and $I_{L \max} = 200 \text{ mA}$:

$$I_R = I_{Z \min} + I_{L \max} = 0 + 200 = 200 \text{ mA}$$

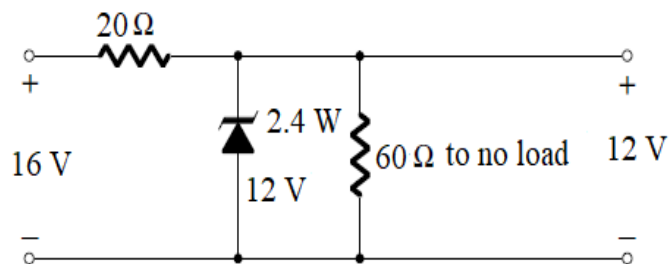
$$V_R = V_i - V_Z = 16 - 12 = 4 \text{ V}$$

$$R = \frac{V_R}{I_R} = \frac{4}{200} = 0.02 \text{ k}\Omega = 20 \Omega$$

$$\text{Maximum load resistance: } R = \frac{V_L}{I_{L \min}} = \frac{12}{0} = \infty \text{ (open circuit } \Rightarrow \text{ no load)}$$

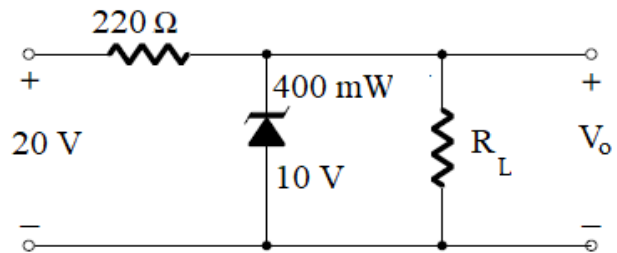
$$\text{Minimum load resistance: } R = \frac{V_L}{I_{L \max}} = \frac{12}{200} = 0.06 \text{ k}\Omega = 60 \Omega$$

$$\text{Maximum power rating: } P_{Z \max} = V_Z \times I_Z = 12 \times 200 \text{ m} = 2400 \text{ mW} = 2.4 \text{ w}$$



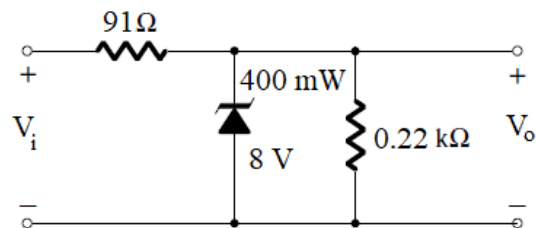
Exercises

1. For the network shown below:



- Determine V_L , I_L , I_Z , and I_R if $R_L = 180 \Omega$
- Repeat part (a) if $R_L = 470 \Omega$
- Determine the value of R_L that will establish maximum power conditions for the Zener diode.
- Determine the minimum value of R_L to ensure that the Zener diode is in the "on" state.

2. For the network shown below. Determine the range of V_i that will maintain V_L at 8 V and not exceed the maximum power rating of the Zener diode.



3. Design a voltage regulator that will maintain an output voltage of 20 V across a 1 kΩ load with an input that will vary between 30 and 50 V.

Answers

- $V_L = 9 \text{ V}$, $I_L = 50 \text{ mA}$ and $I_Z = 0$
 - $V_L = 10 \text{ V}$, $I_L = 21.27 \text{ mA}$ and $I_Z = 24.18 \text{ mA}$
 - $R_L = 1.83 \text{ k}\Omega$
 - $R_L = 220 \Omega$
- 11.31 V to 15.86 V
- $V_Z = 20 \text{ V}$, $R = 0.5 \text{ k}\Omega$ and $P_{Z \text{ mzx}} = 800 \text{ mW}$